FDOPA (Fluorodopa) PET scan is a specialized imaging technique used to visualize how the brain produces and uses dopamine. It is a critical tool for diagnosing and understanding the progression of Parkinson's disease.

HOW THE SCAN WORKS

- The tracer: A chemical tracer called FDOPA is injected into the patient.
- Dopamine Visualization: This tracer mimics L-dopa (a precursor to dopamine), allowing the PET scan to show actual dopamine synthesis and activity within the brain.
- Healthy Activity: In a healthy brain, the tracer "lights up" in bright red and yellow colors in the striatum (the area responsible for movement).

FDOPA Scans in Parkinson's Disease

In patients with Parkinson's, the scan reveals physical changes in the brain's chemistry.

- Reduced Uptake: The bright yellow/red regions fade to blue or disappear entirely.

Cell Loss: This color changes indicates the neurons responsible for producing dopamine are either dying or functioning poorly.

Assymetry: The loss of dopamine activity often begins on one side of brain, which usually corresponds to the side of body where the patient first noticed tremors or stiffness.

Clinical Importance:

1. Early Diagnosis - Helps doctors detect Parkinson's by seeing chemical changes before physical symptoms become severe.
2. Mapping Movement: Identifies issues in the striatum, which recieves the dopamine necessary for movement control.
3. Tracking Progression: Allows medical professionals to monitor how the disease changes over time by observing the fading intensity of tracer.

Color Spectrum of Dopamine

- **Bright Red and White / Yellow**: These represent the highest levels of dopamine synthesis and uptake. In a healthy brain, these colors appear as vibrant, bean-shaped regions in the striatum.
- **Green & Yellow**: These are "middle-ground" colors indicating moderate activity.
- **Blue and Purple**: These represent low or reduced activity. As Parkinson's progresses, the bright red fade into these cooler colors.
- **Black / Grey**: These indicates background tissue where there is no tracer uptake.

Visualizing the Difference:

1. **Normal Brain** ⇒ Dominant color in striatum: Bright Red / Yellow.
   Neurons are healthy & producing dopamine normally.

2. **Parkinson's Brain** ⇒ Faded Blue / Green.
   - Neurons are dying or malfunctioning, leading to less dopamine.

Why color fades:

In Parkinson's Disease, the loss of dopamine usually starts on one side of brain. On the scan, you will often see one hemisphere still showing some orange, or yellow, while the other side has faded almost entirely to blue. This "color loss" typically matches the side of the body where the patient feels the most stiffness or tremors.

Q1. In the FDOPA, scan shown above what does bright, red/yellow region represent?

→ The bright red and yellow regions represent high dopamine activity and synthesis. Specifically, they show where healthy neurons are actively taking up the FDOPA tracer in striatum.

Q2. Compare the 'Normal' and 'Parkinson' brain images. What major difference do you see?

→ The primary difference is intensity and color of the tracer.

- Normal Brain: Shows large, bright red and yellow areas in both hemispheres, indicating high dopamine activity.
- Parkinson's Brain: These bright colors fade to blue, indicating a significant reduction in dopamine uptake because the neurons are dying or malfunctioning.

Q3. Why is dopamine important for movement control?

→ Dopamine acts as a chemical messenger that allows the brain to communicate with muscle. It is essential for coordinating smooth, purposeful movements; without enough dopamine, the brain cannot properly signal the body, leading to the tremors and stiffness seen in Parkinson's.

Q4. Match the following:

1. Substantia nigra → Produces dopamine.
2. Striatum → Receives dopamine for movement control.
3. FDOPA → Tracer used to visualize dopamine activity

Q5. Imagine you are explaining this scan to a patient in simple language.

→ "Think of this scan like a battery check for your brain's movement centre. The bright 'glowing' spot in a healthy scan show that your brain is making plenty of dopamine to keep your movements smooth. In this scan, those spots have dimmed or changed color, which tells us that the cells responsible for making the 'fuel' are struggling, which is why you are experiencing difficulty with movements."

Q6. (Bonus) How can such scans help doctors to detect Parkinson's early or track its progressions?

→ These scans allow doctors to see physical changes in brain chemistry before physical symptoms might be fully apparent. By comparing scans over time, decades / doctors can track how quickly dopamine activity is decreasing, helping them quickly/~~qui~~ adjust treatements and monitors how the disease is progressing in a specific patient.